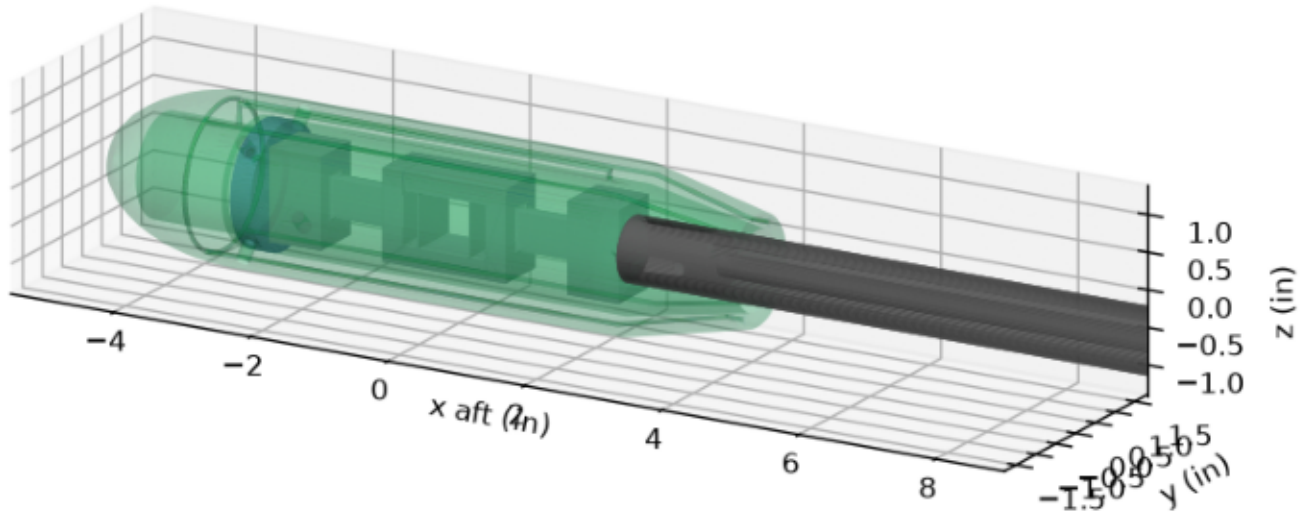


FLIGHT ASSEMBLY (CAD RENDER, NTS)

Flight assembly: instrumented 2.25 in pod (transparent), puck, 6-component balance, sting



NOTES:

1. GAUGE AND WIRE THE BALANCE (24 GAUGES, 6 BRIDGES) BEFORE ASSEMBLY; SEE THE GAUGE LAYOUT DRAWING.
2. PUCK TO BALANCE FRONT FACE: 2 DOWELS + 2 SCREWS. STING ON THE SPIGOT: CROSS PIN + 2 SET SCREWS.
3. WIRES RUN AFT THROUGH THE BALANCE AND THE STING BORE TO J0 AT THE ROLL HOUSING.
4. THE MODEL POD IS 3D PRINTED FROM THE STEP FILE (NO MACHINING DRAWING).

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

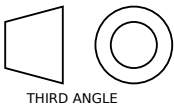
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

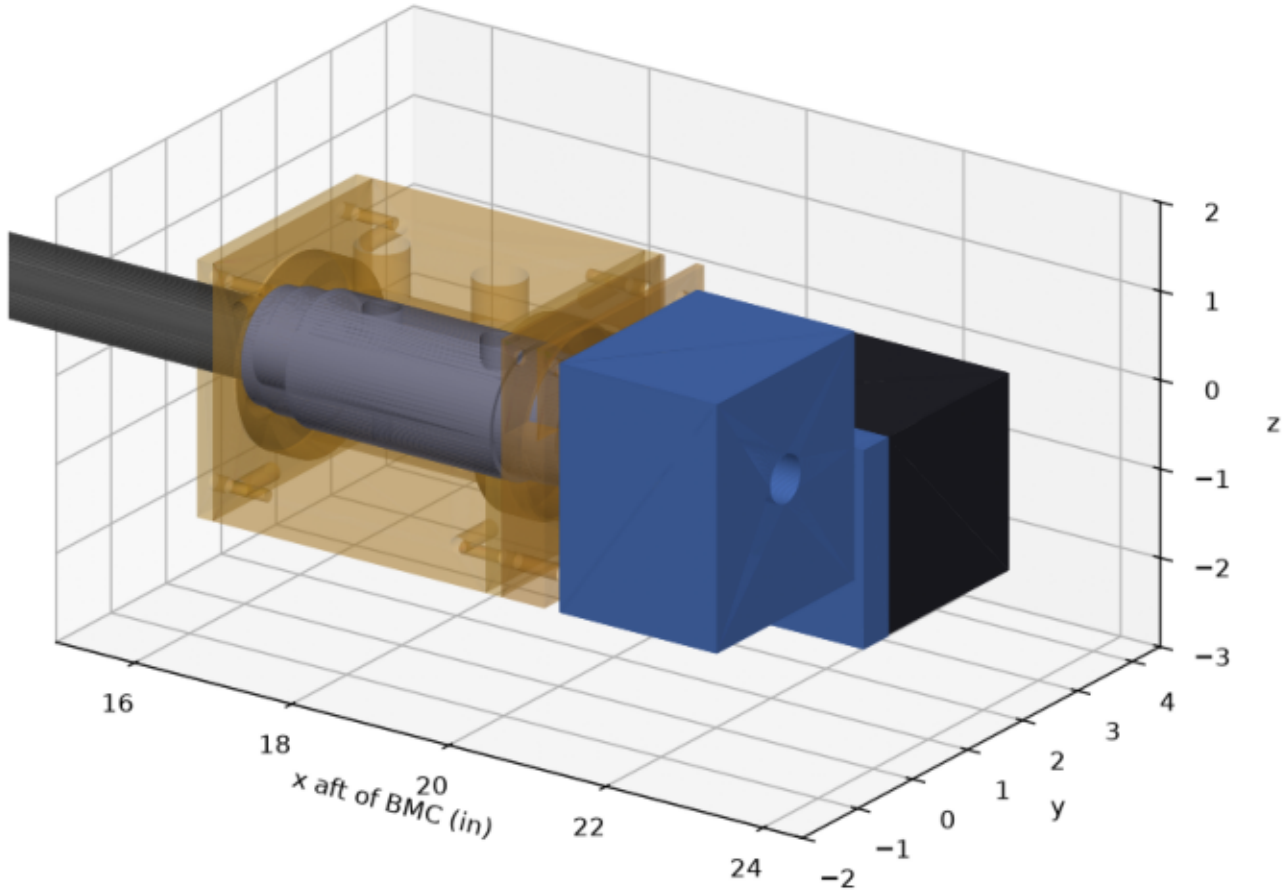
ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-001	Balance, 6-component (gauged)	1	7075-T651
2	WT-6C-002	Puck, model interface	1	6061-T6
3	WT-6C-003	Sting	1	4140 28-32 HRC
4	POD-2.25	Pod, reference model (STEP)	1	3D printed
5	-	Strain gauge 350 ohm, CEA-13 class	24	Vishay / bought
6	-	Dowel pin 1/8 x 1/2	2	bought
7	-	SHCS #10-32 x 5/8 (puck)	2	bought
8	-	Cross pin 3/16 dowel (sting)	1	bought
9	-	Set screw 1/4-28 cup point	2	bought
10	-	SHCS #8-32 x 3/8 (pod)	4	bought
11	-	Wire 32 AWG PTFE + J0 DC-37	1	see Electronics

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ASSEMBLY, FLIGHT (BALANCE + STING + MODEL)							
MATERIAL				FINISH			
SEE PARTS LIST				SEE PART DRAWINGS			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-100		A	B	NTS		-	1 OF 35

ROLL SPINDLE AT ROLL 0 (CAD RENDER, NTS)

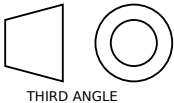
Roll spindle at roll +0 deg: housing/caps/adapter transparent, NMRV030 50:1 + NEMA 23, AS5048A encoder



NOTES:

- PRESS BOTH BEARINGS ONTO THE SPINDLE SEATS; FIT INTO THE HOUSING; CLAMP WITH THE CAPS. SHIM THE REAR CAP FOR .002-.004 AXIAL FLOAT; TIGHTEN THE KM7 LOCKNUT.
- STING FLATS IN THE D-D BORE; 2 BOLTS THROUGH THE TOP WINDOWS AT ROLL 0.
- AS5048A CHIP ON THE SPINDLE AXIS WITHIN 0.25 MM, 1.5 MM AIR GAP.
- BALANCE CABLE EXITS AT X = 20.53: LEAVE A SERVICE LOOP FOR 270 DEG OF TWIST (CALIBRATION MODE).

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X ±.03 .XX ±.010
.XXX ±.005 .XXXX ±.0005
ANGLES ±0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 µin Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

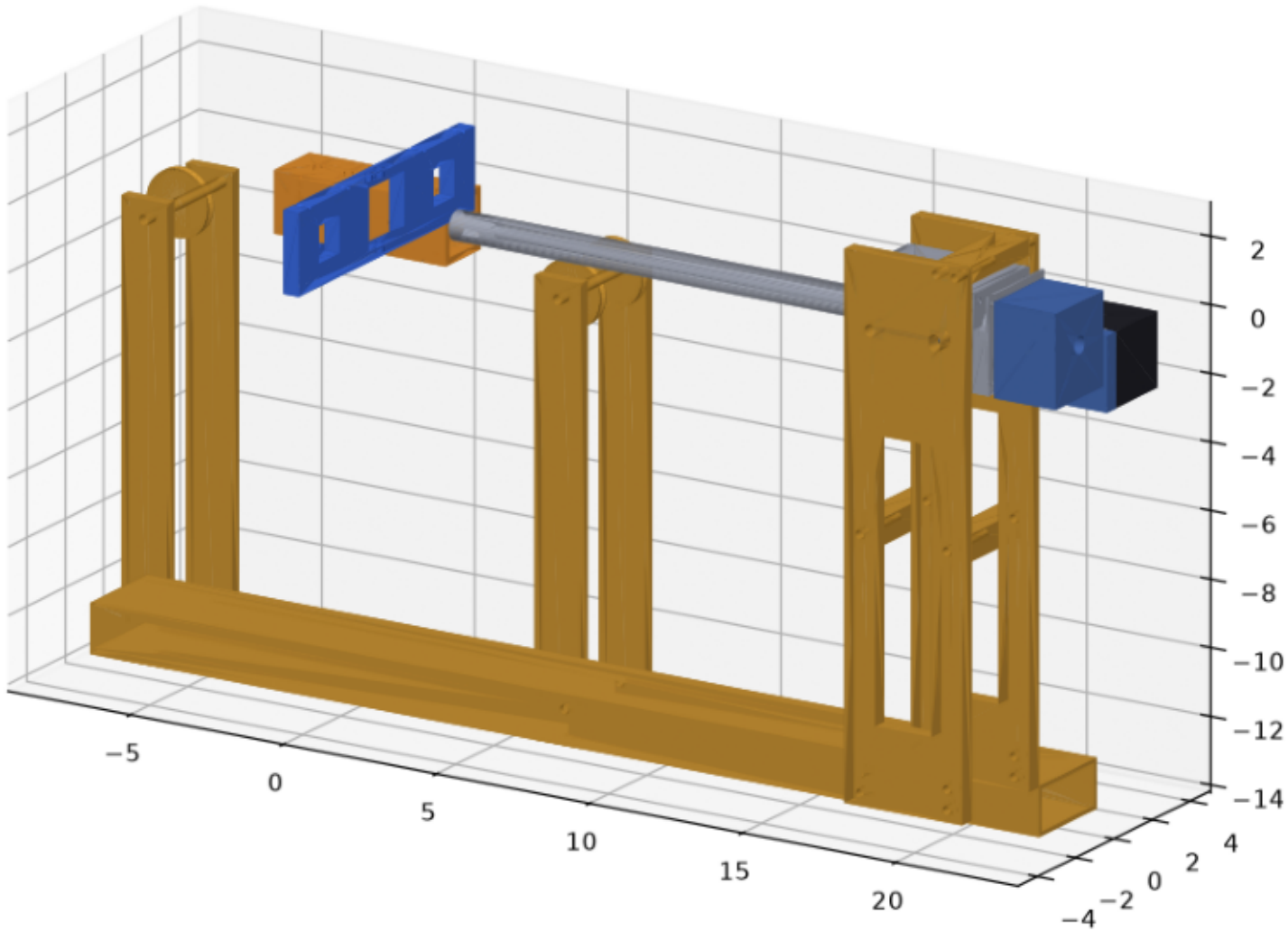
ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-201	Spindle, roll	1	4140
2	WT-6C-202	Housing, roll spindle	1	6061-T6
3	WT-6C-203-1	Cap, bearing, front	1	6061-T6
4	WT-6C-203-2	Cap, bearing, rear	1	6061-T6
5	WT-6C-204	Adapter, gearbox	1	6061-T6
6	WT-6C-205	Bracket, roll encoder	1	6061-T6
7	-	Bearing 6907-2RS (35x55x10)	2	bought
8	-	Locknut KM7 + lock washer MB7	1	bought
9	-	Worm gearbox NMRV030 50:1, NEMA 23 input	1	bought
10	-	Stepper motor NEMA 23	1	bought
11	-	Encoder AS5048A board + 6x2.5 diametric magnet	1	bought
12	-	SHCS #8-32 (caps, adapter)	16	bought
13	-	SHCS M6 (gearbox), M3 standoffs (encoder)	1	bought
14	-	Shims .002/.005 (rear cap float)	1	bought

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, ROLL SPINDLE								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-200		A	B	NTS		-		2 OF 35

CALIBRATION ON THE ROLL SPINDLE (CAD RENDER, NTS)

6-component calibration on the roll spindle: square sleeve, roll arm, yoke at x = -2, drag/thrust pulleys



NOTES:

1. THE BALANCE IS CALIBRATED ON THE REAL ROLL SPINDLE: THE HOUSING PIVOTS ON ITS FRONT STUD BETWEEN THE HEAD PLATES; THE REAR STUD SITS IN THE LEVELING SLOT; RE-LEVEL WITH THE JACK SCREW AFTER EVERY LOAD.
2. SLEEVE REPLACES THE PUCK. YOKE AT ROLL 0/90/180/270, STIRRUP AT 45/135/315, ROLL ARM AT 0/180.
3. BENCH MODE ROLL RANGE -90..+180 DEG. SCHEDULE AND FIT: analysis/balance_calibration_fit_6c.py.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

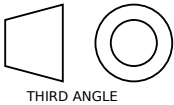
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



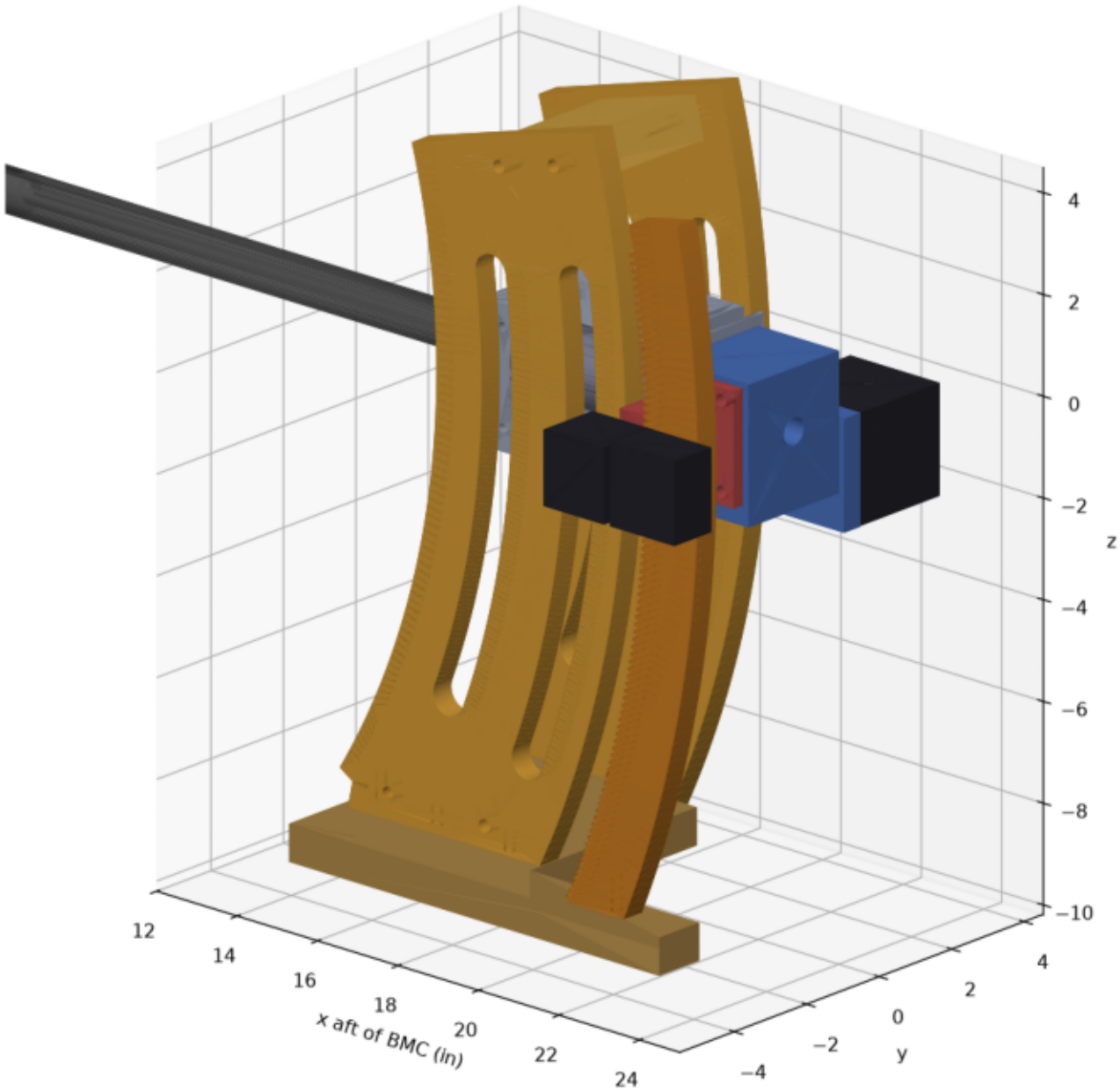
PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-301	Sleeve, calibration	1	6061-T6
2	WT-6C-302	Arm, roll calibration	1	6061-T6
3	WT-6C-303	Stirrup, 45 deg	1	6061-T6
4	WT-6C-304	Yoke, calibration	1	6061-T6
5	WT-6C-305	Plate, head	2	6061-T6
6	WT-6C-306-1	Bridge, jack	1	6061-T6
7	WT-6C-306-2	Spacer, cross	2	6061-T6
8	WT-6C-307	Tube, base	1	6061-T6 tube
9	WT-6C-308	Upright, pulley	4	6061-T6
10	WT-6C-309	Axle, pulley	2	416 SS ground
11	WT-6C-310	Pulley, cable, 2 in	3	6061-T6
12	-	Bearing R5-2Z (5/16 x 1/2)	3	bought
13	-	Shaft collar 5/16	4	bought
14	WT-6C-200	Roll spindle assy (with balance + sting)	1	see WT-6C-200
15	-	Stud 3/8 plain + nut (housing pivot/level)	4	bought
16	-	Jack screw 1/4-28 brass tip	1	bought
17	-	Ground pin 1/4 (load pin), dowels 1/4	3	bought
18	-	Cable 1/16 SS, eyebolt 1/4-20, hooks	1	bought
19	-	Dead weights, calibrated (Class F)	1	bought
20	-	SHCS 1/4-20 + nuts	16	bought

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ASSEMBLY, 6-COMPONENT CALIBRATION							
MATERIAL				FINISH			
SEE PARTS LIST				SEE PART DRAWINGS			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-300		A	B	NTS		-	3 OF 35

PITCH SECTOR WITH ROLL SPINDLE (CAD RENDER, NTS)

Motorized pitch sector at alpha 0: braced cheeks with arc tracks, rack plate + pinion drive (-y side), roll spindle carried on 4 cam followers



NOTES:

1. BOLT THE CHEEKS, TIE AND SPACERS TOGETHER ON THE BASE PLATE FIRST; CHECK THE CHEEK SPACING (3.000).
2. INSTALL THE CAM FOLLOWERS IN THE ROLL HOUSING; LOWER THE ROLL ASSEMBLY INTO THE ARC SLOTS.
3. SET THE RACK-PINION BACKLASH .004-.008 IN; GRAVITY KEEPS THE SAME FLANK LOADED.
4. RANGE -6..+20 DEG; PITCH RATE LIMITED TO 2 DEG/S.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

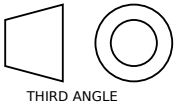
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

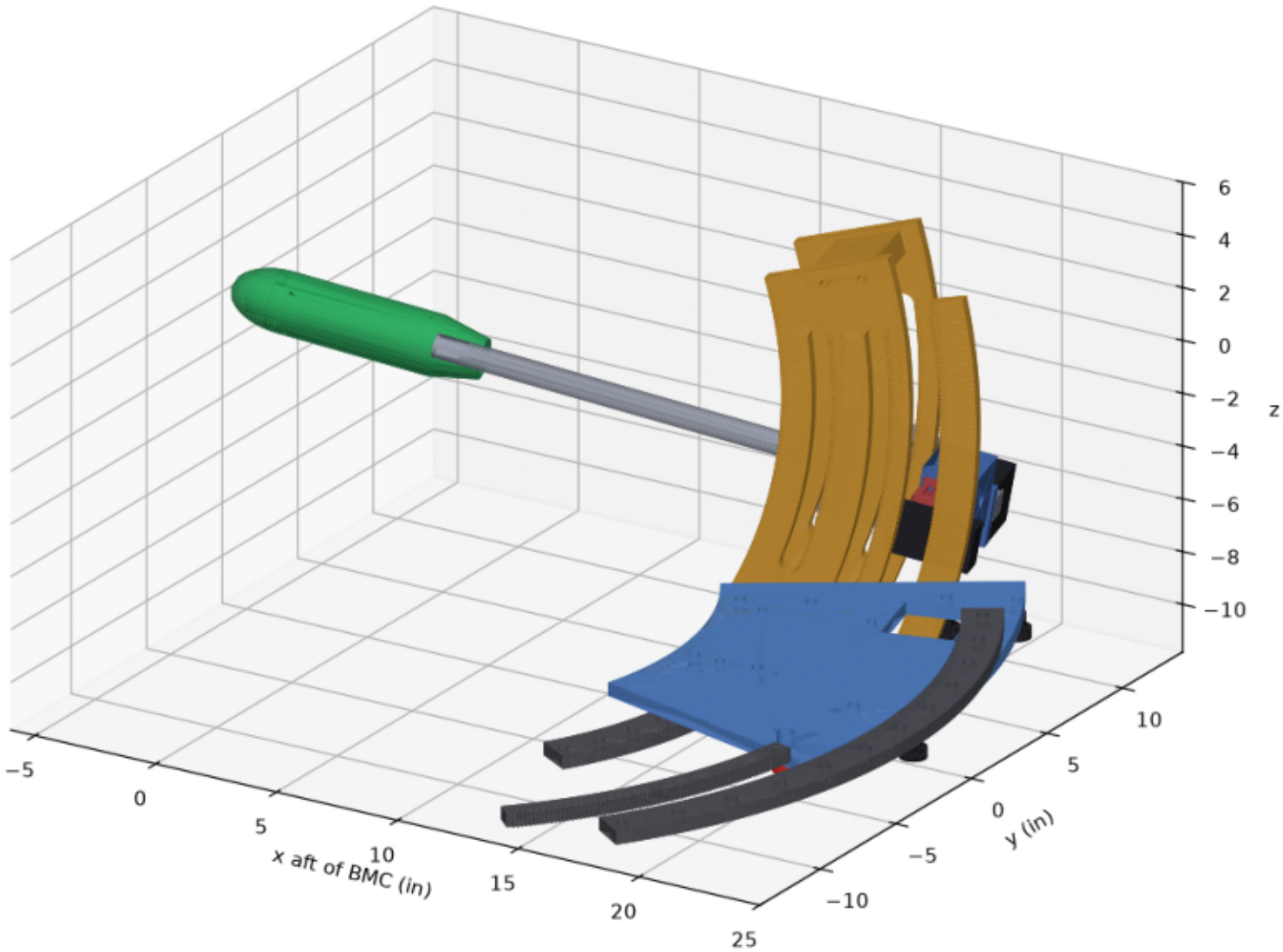
ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-401	Cheek, pitch sector	2	6061-T6
2	WT-6C-402	Rack plate, pitch	1	6061-T6
3	WT-6C-403	Plate, base, pitch	1	6061-T6
4	WT-6C-404-1	Tie, top	1	6061-T6
5	WT-6C-404-2/3	Spacers, fwd / aft	2	6061-T6
6	WT-6C-405	Bracket, pitch drive	1	6061-T6
7	WT-6C-406	Pinion shaft (+ stock m1.25 20T gear)	1	4140 + bought gear
8	-	Bearing 608-2RS	2	bought
9	-	Circlip DIN 471-8	1	bought
10	-	Cam follower 3/4 in, 3/8-24 stud	4	bought
11	-	Worm gearmotor 40:1 + NEMA 17	1	bought
12	WT-6C-200	Roll spindle assy	1	see WT-6C-200
13	WT-6C-100	Flight assy (balance, sting, model)	1	see WT-6C-100
14	-	SHCS 1/4-20 (cheeks, tie, spacers, rack)	14	bought
15	-	SHCS M6 (drive bracket)	4	bought
16	-	Limit switches + inclinometer	1	see Electronics

NEXT ASSY: WT-6C-500 YAW CARRIAGE ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, MOTORIZED PITCH SECTOR								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.	REV	SIZE	SCALE	WEIGHT	SHEET			
WT-6C-400	A	B	NTS	-	4 OF 35			

COMPLETE MOTION SYSTEM ON THE YAW CARRIAGE (CAD RENDER, NTS)

Complete motion system at yaw +15, pitch +10: floor rails -> yaw carriage -> pitch sector -> roll spindle -> sting -> model



NOTES:

1. PRELIMINARY: RAIL POSITIONS AND FLOOR HOLES DEPEND ON THE DOWNSTREAM BAY FLOOR (TO CONFIRM).
2. INSTALL THE RAILS CONCENTRIC WITH THE YAW AXIS (THROUGH THE BMC) USING A RADIUS BAR ON A PIN.
3. PRELOAD EACH TRUCK WITH ITS ECCENTRIC SPACER UNTIL THE WHEELS TURN WITH LIGHT DRAG.
4. RANGE +-15 DEG YAW; THE PLATE REACHES ABOUT 2.3 IN UNDER THE TEST-SECTION EXIT AT FULL YAW.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

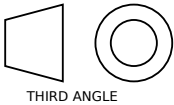
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL

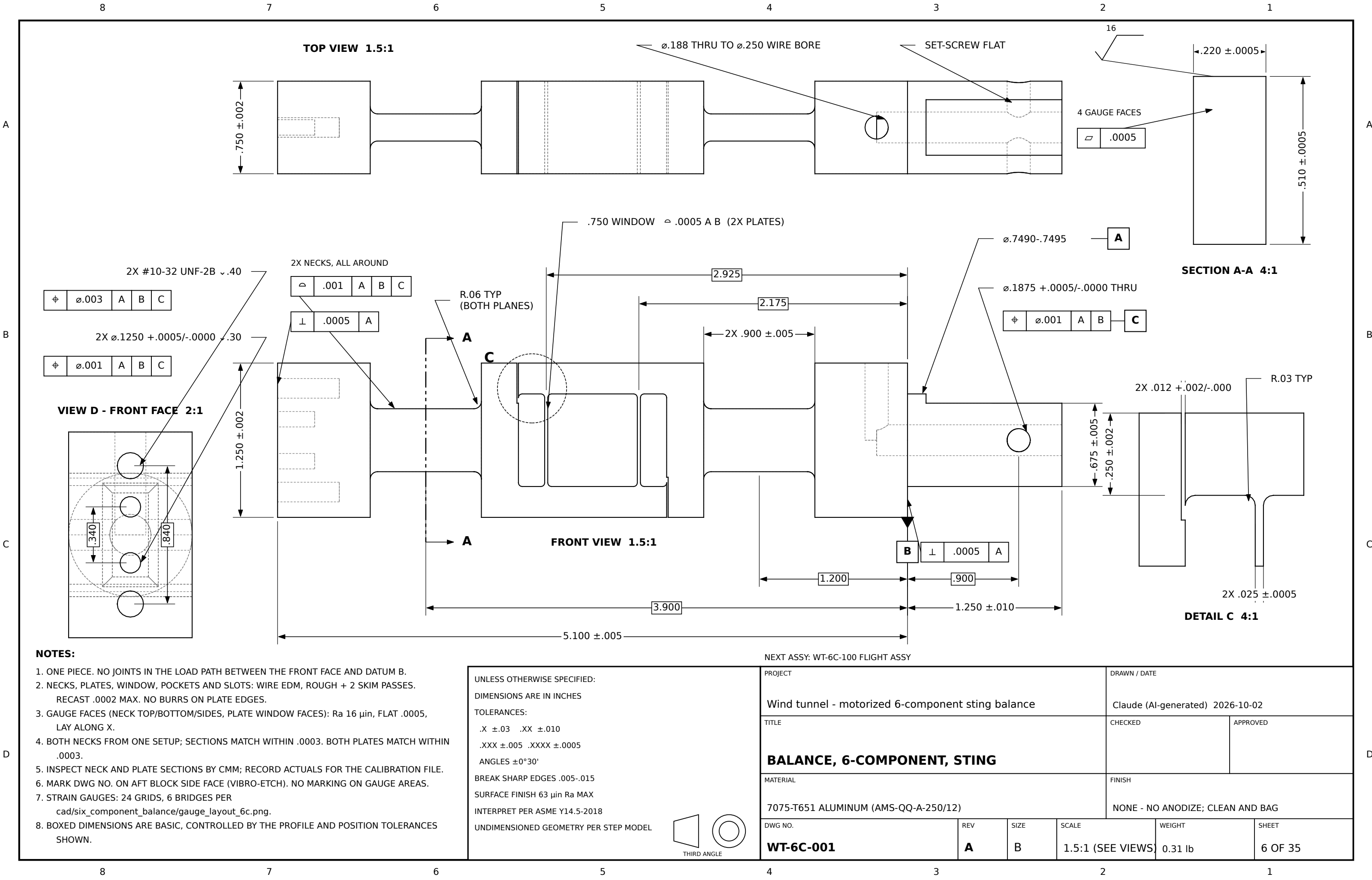


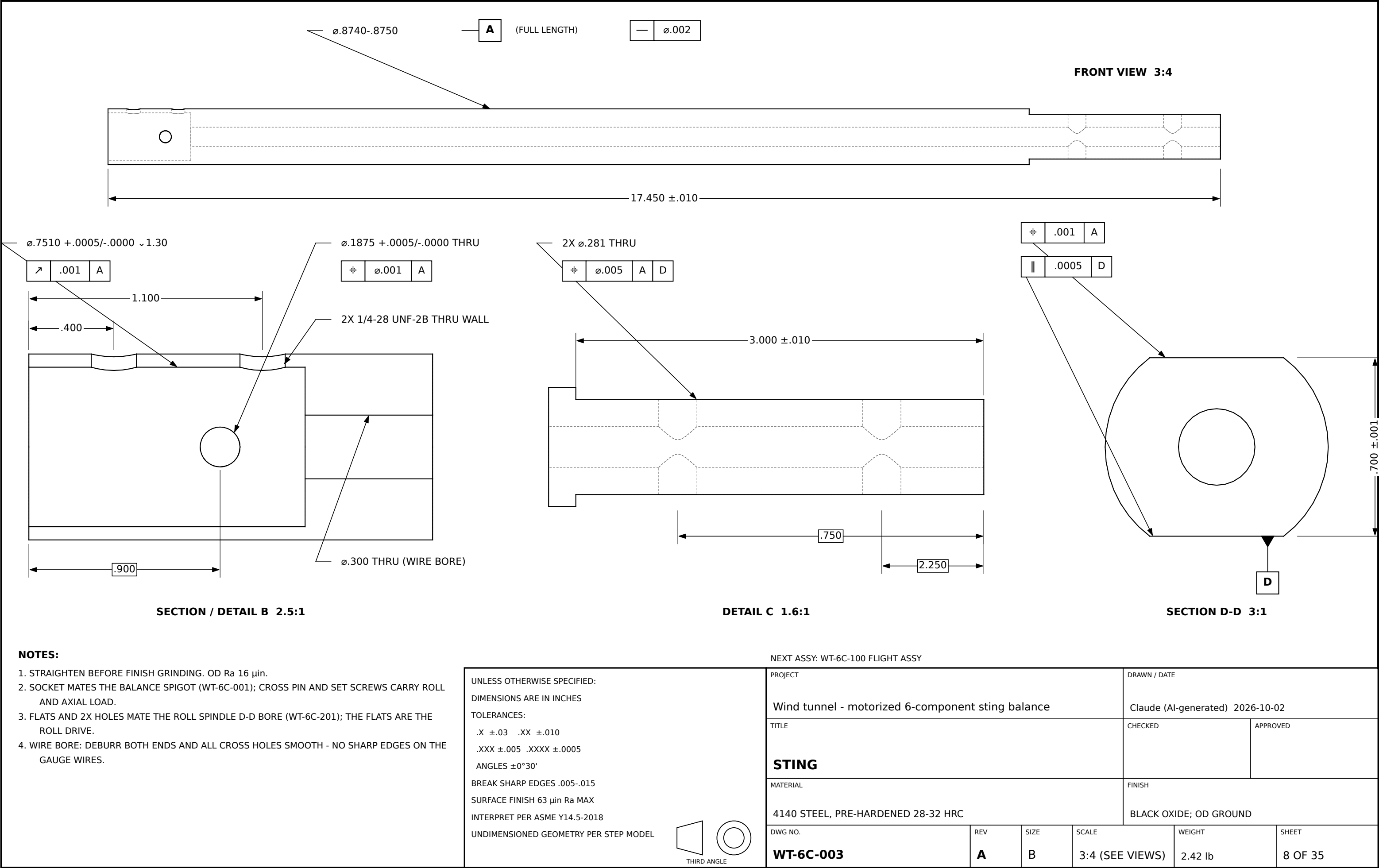
PRELIMINARY - TO CONFIRM

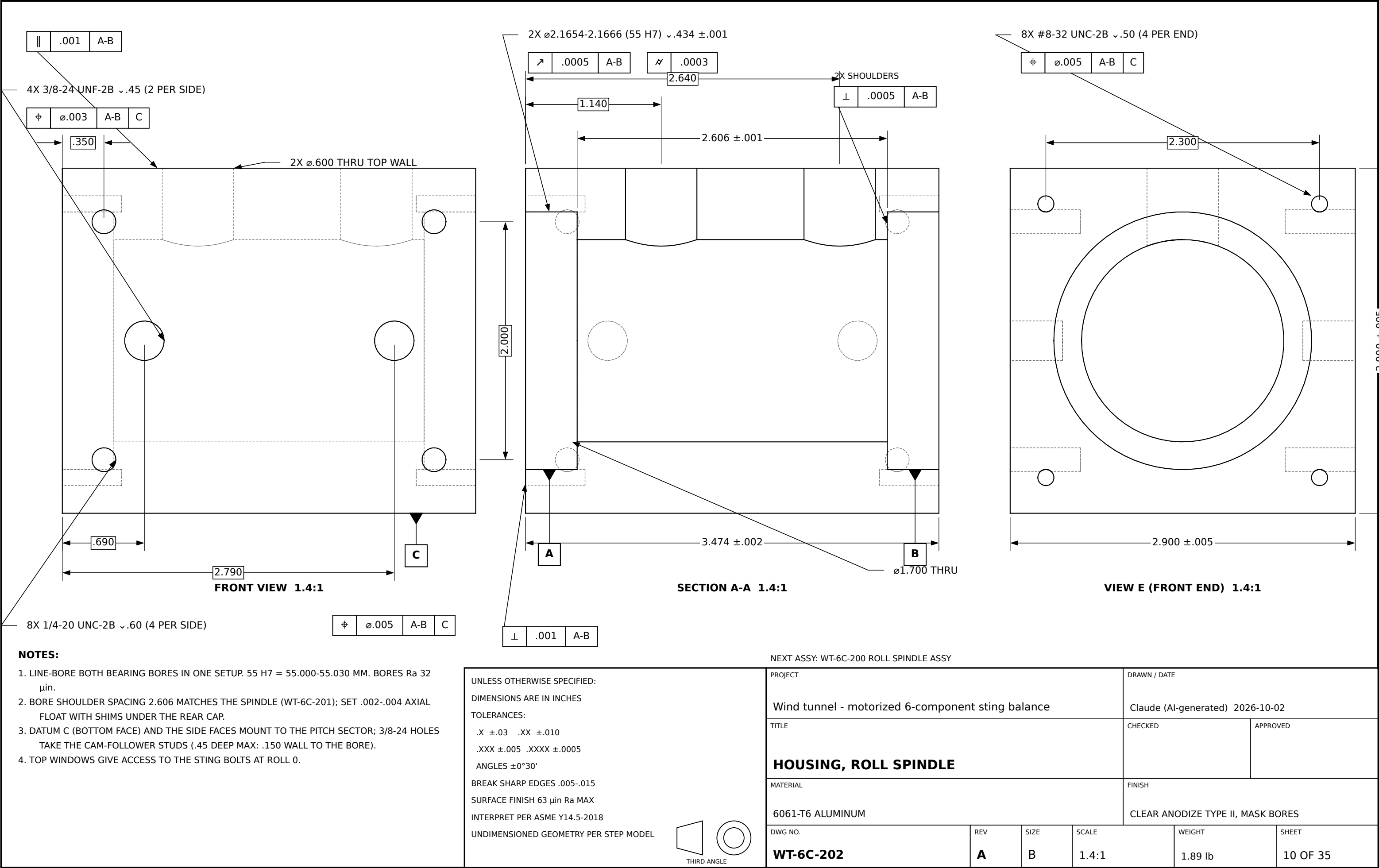
PARTS LIST

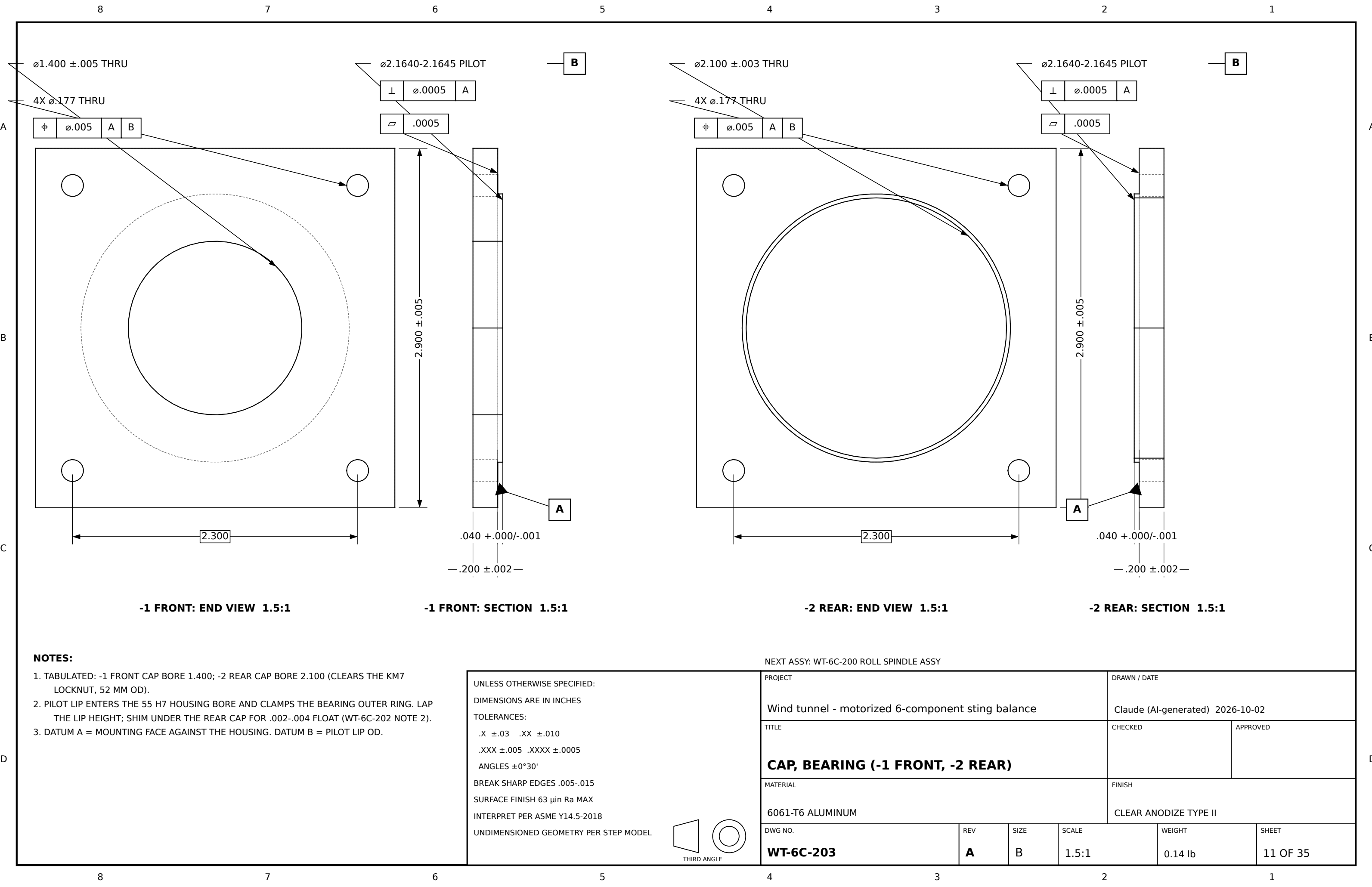
ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-501	Rail, yaw, inner	1	6061-T6
2	WT-6C-502	Rail, yaw, outer	1	6061-T6
3	WT-6C-503	Rack, yaw, floor	1	6061-T6
4	WT-6C-504	Plate, yaw carriage	1	6061-T6
5	WT-6C-505	Pinion, yaw (modified stock gear)	1	steel, bought
6	WT-6C-506-1	Spacer, wheel, plain	4	6061-T6
7	WT-6C-506-2	Spacer, wheel, eccentric	4	6061-T6
8	-	V-groove wheel 1.00 OD, 90 deg	8	bought
9	-	Wheel stud / shoulder bolt + nut	8	bought
10	-	Worm gearmotor 30:1 + NEMA 17	1	bought
11	WT-6C-400	Pitch sector assy	1	see WT-6C-400
12	-	FHCS 3/8 (rails, rack to floor)	26	bought
13	-	SHCS 3/8-16 (sector base to carriage)	4	bought
14	-	Home / limit switches	1	see Electronics

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ASSEMBLY, YAW CARRIAGE							
MATERIAL				FINISH			
SEE PARTS LIST				SEE PART DRAWINGS			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-500		A	B	NTS		-	5 OF 35





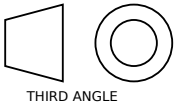




NOTES:

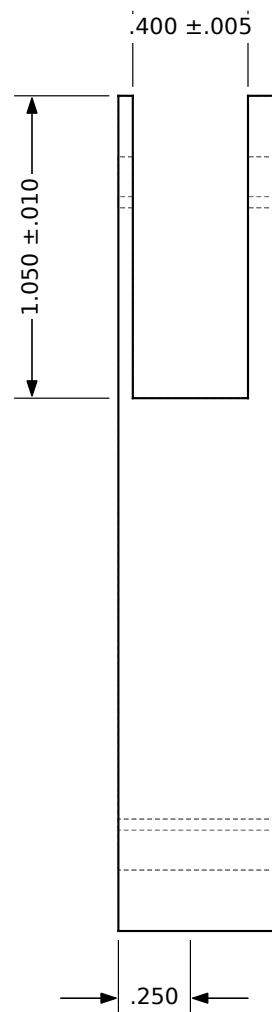
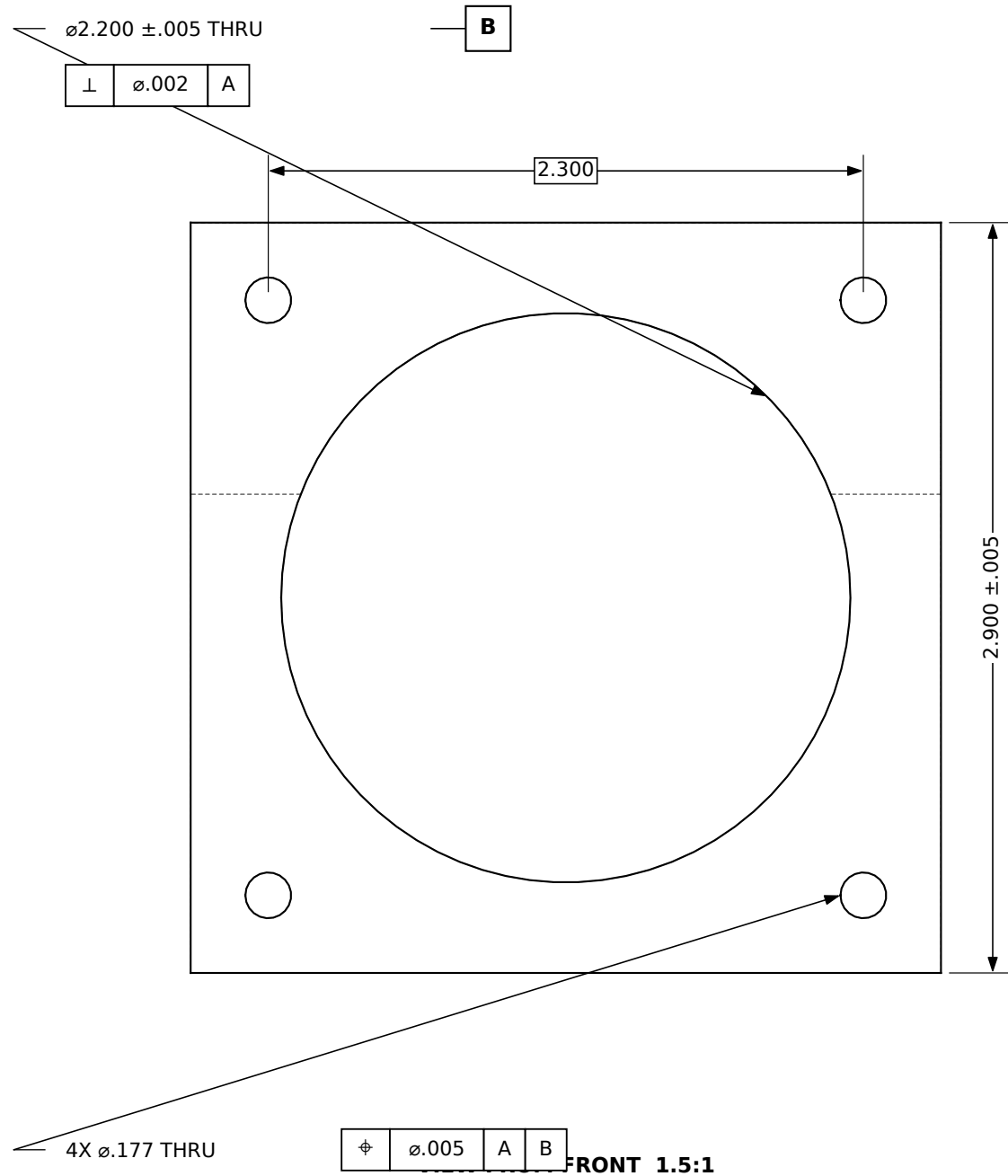
- TABULATED: -1 FRONT CAP BORE 1.400; -2 REAR CAP BORE 2.100 (CLEARS THE KM7 LOCKNUT, 52 MM OD).
- PILOT LIP ENTERS THE 55 H7 HOUSING BORE AND CLAMPS THE BEARING OUTER RING. LAP THE LIP HEIGHT; SHIM UNDER THE REAR CAP FOR .002-.004 FLOAT (WT-6C-202 NOTE 2).
- DATUM A = MOUNTING FACE AGAINST THE HOUSING. DATUM B = PILOT LIP OD.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X ±.03 .XX ±.010
.XXX ±.005 .XXXX ±.0005
ANGLES ±0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 µin Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL

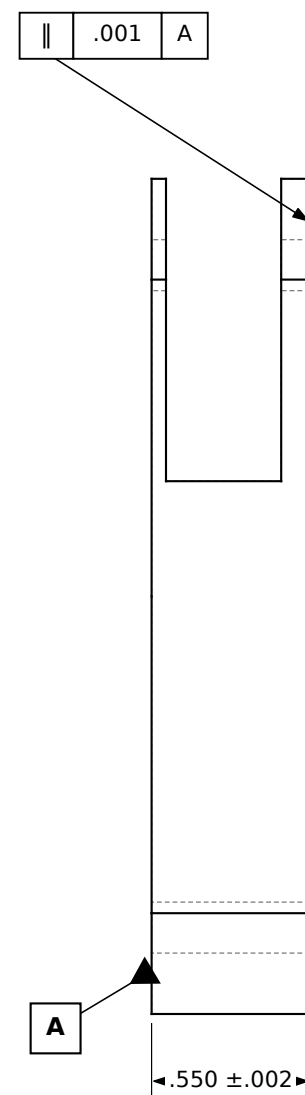


NEXT ASSY: WT-6C-200 ROLL SPINDLE ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
CAP, BEARING (-1 FRONT, -2 REAR)							
MATERIAL				FINISH			
6061-T6 ALUMINUM				CLEAR ANODIZE TYPE II			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-203		A	B	1.5:1		0.14 lb	11 OF 35



FRONT VIEW 1.5:1



SECTION A-A 1.5:1

NOTES:

1. AFT FACE: DRILL AND TAP TO SUIT THE NMRV030 OUTPUT-FLANGE PATTERN FROM THE VENDOR DRAWING; LOCATE ON DATUM B SO THE GEARBOX OUTPUT IS COAXIAL WITH THE SPINDLE WITHIN .002.
2. WIRE WINDOW LETS THE SPINDLE WIRE EXIT SWEEP +-90 DEG OF ROLL; DEBURR SMOOTH.
3. DATUM A = FACE AGAINST THE REAR CAP. DATUM B = BORE.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X $\pm .03$	.XX $\pm .010$
--------------	----------------

.XXX ±.005 .XXXX ±.0005

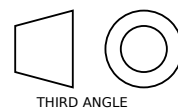
ANGLES $\pm 0^{\circ}30'$

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 μin Ra MAX

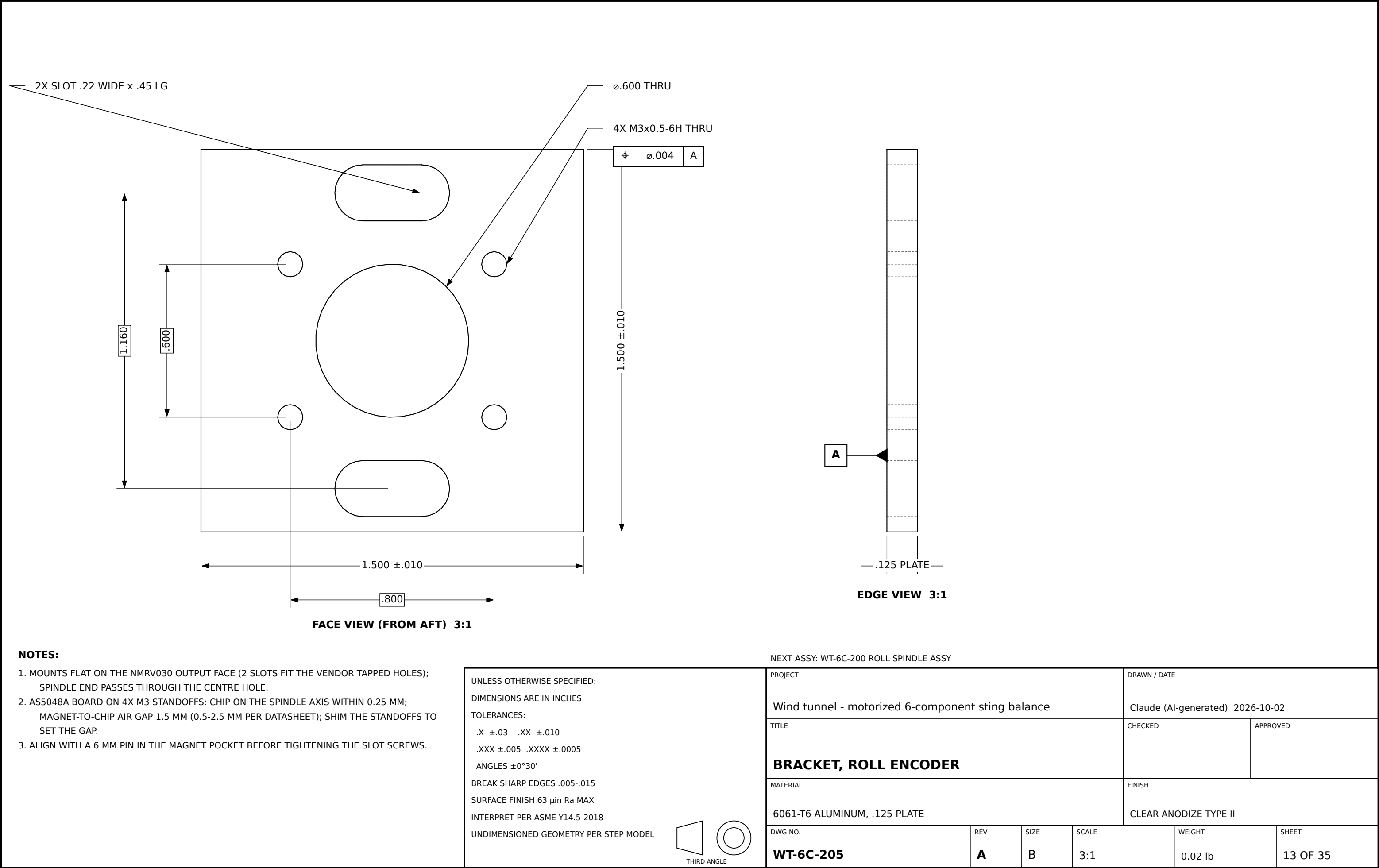
INTERPRET PER ASME Y14.5-2018

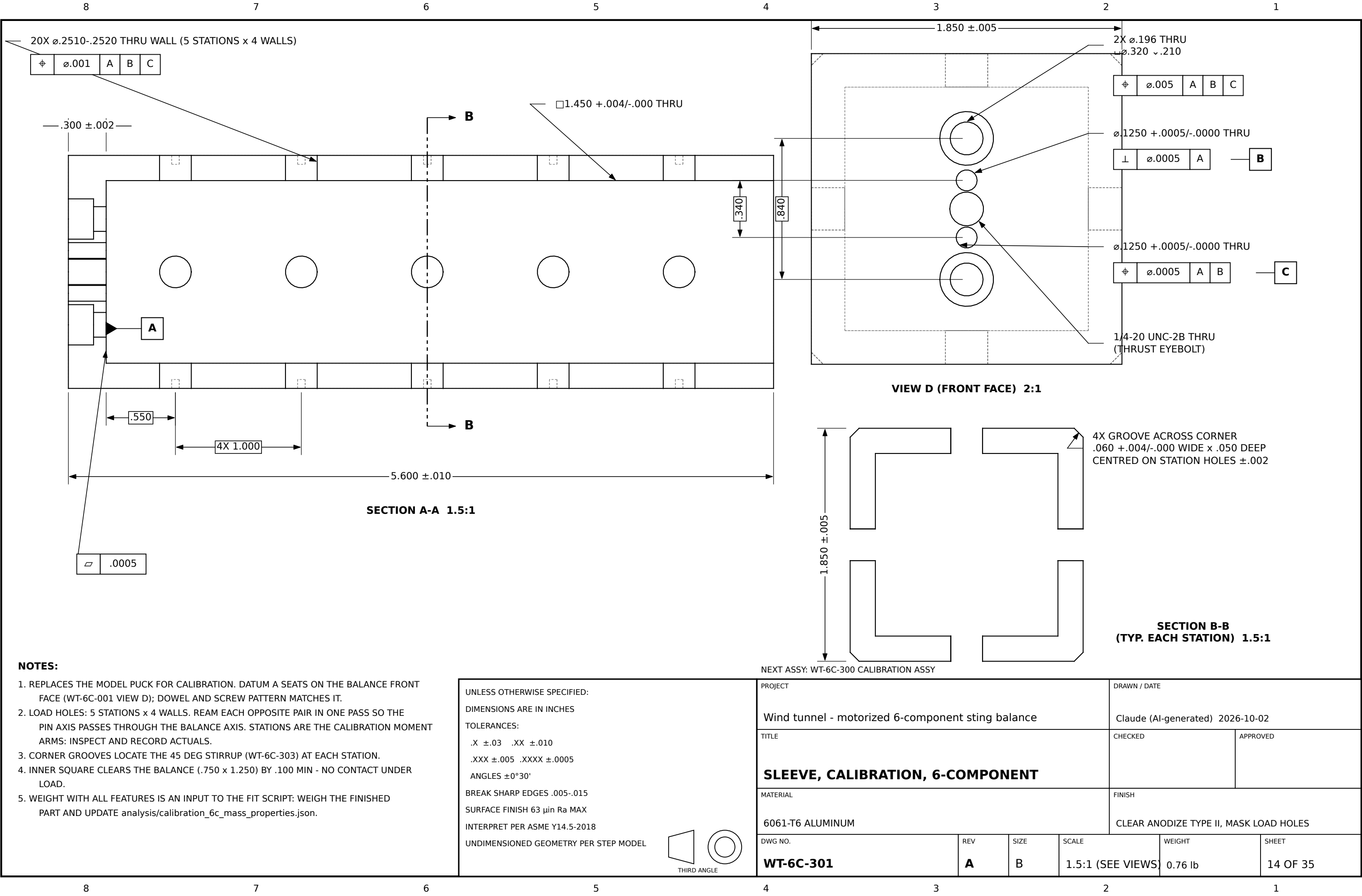
UNDIMENSIONED GEOMETRY PER STEP MODEL

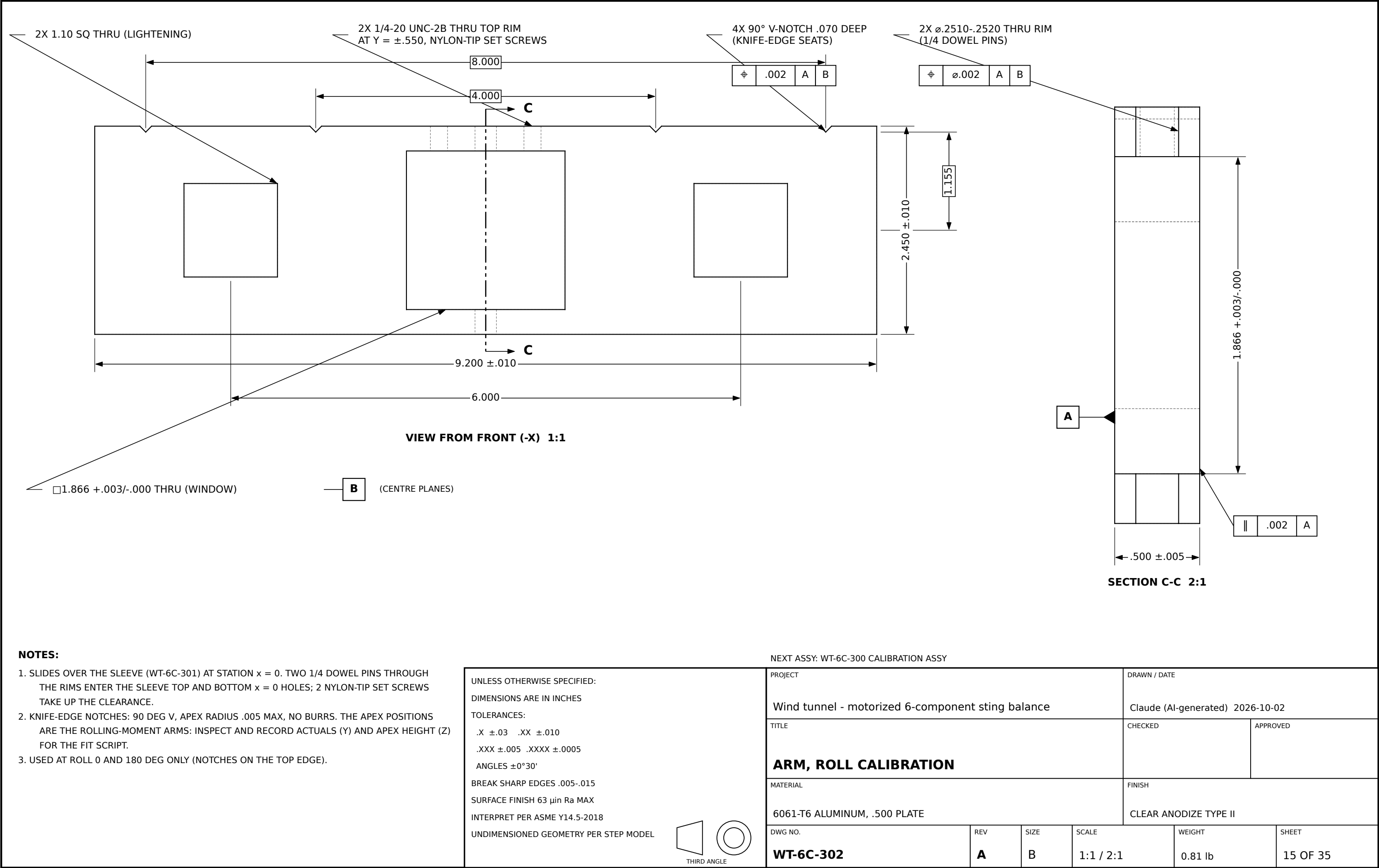


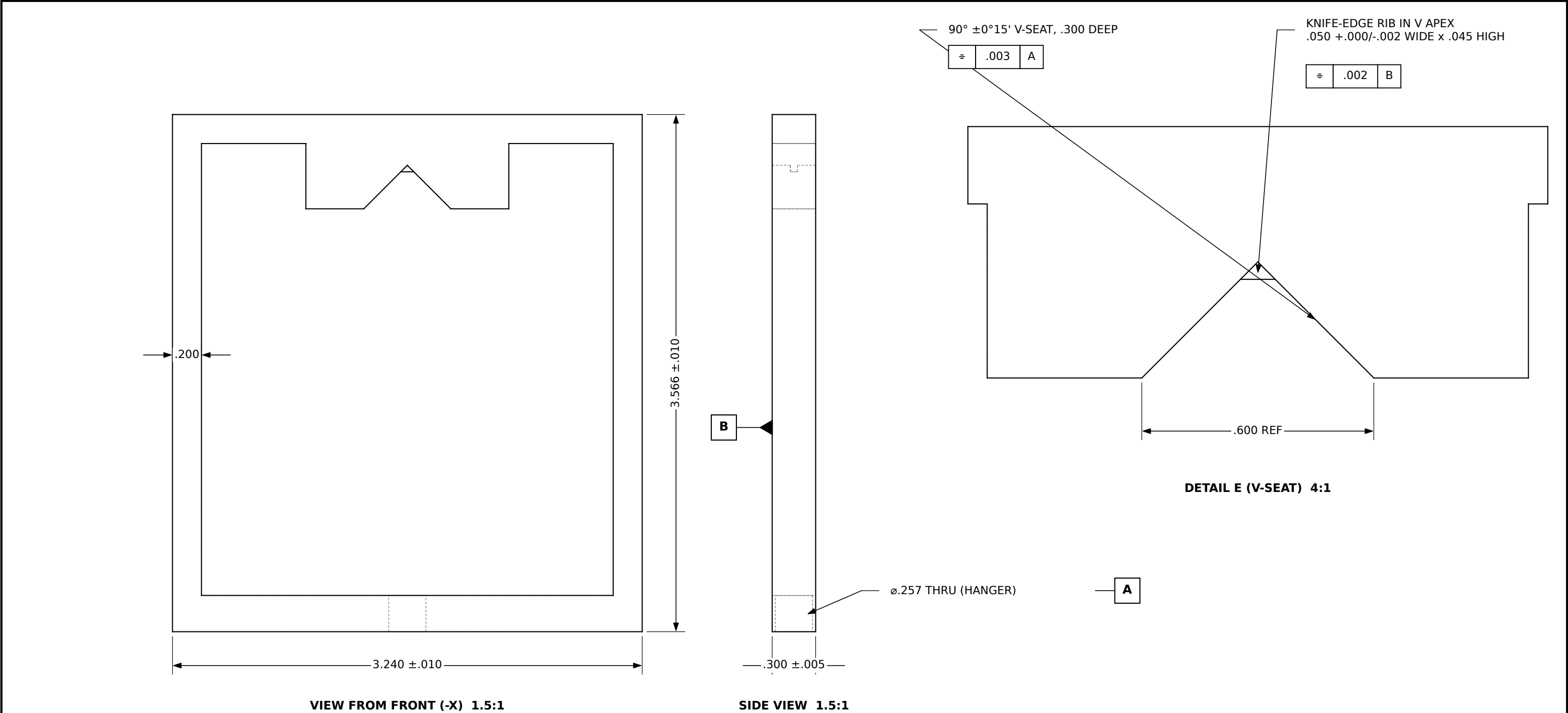
NEXT ASSY: WT-6C-200 ROLL SPINDLE ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ADAPTER, GEARBOX							
MATERIAL				FINISH			
6061-T6 ALUMINUM				CLEAR ANODIZE TYPE II			
DWG NO.		REV	SIZE	SCALE	WEIGHT		SHEET
WT-6C-204		A	B	1.5:1	0.17 lb		12 OF 35



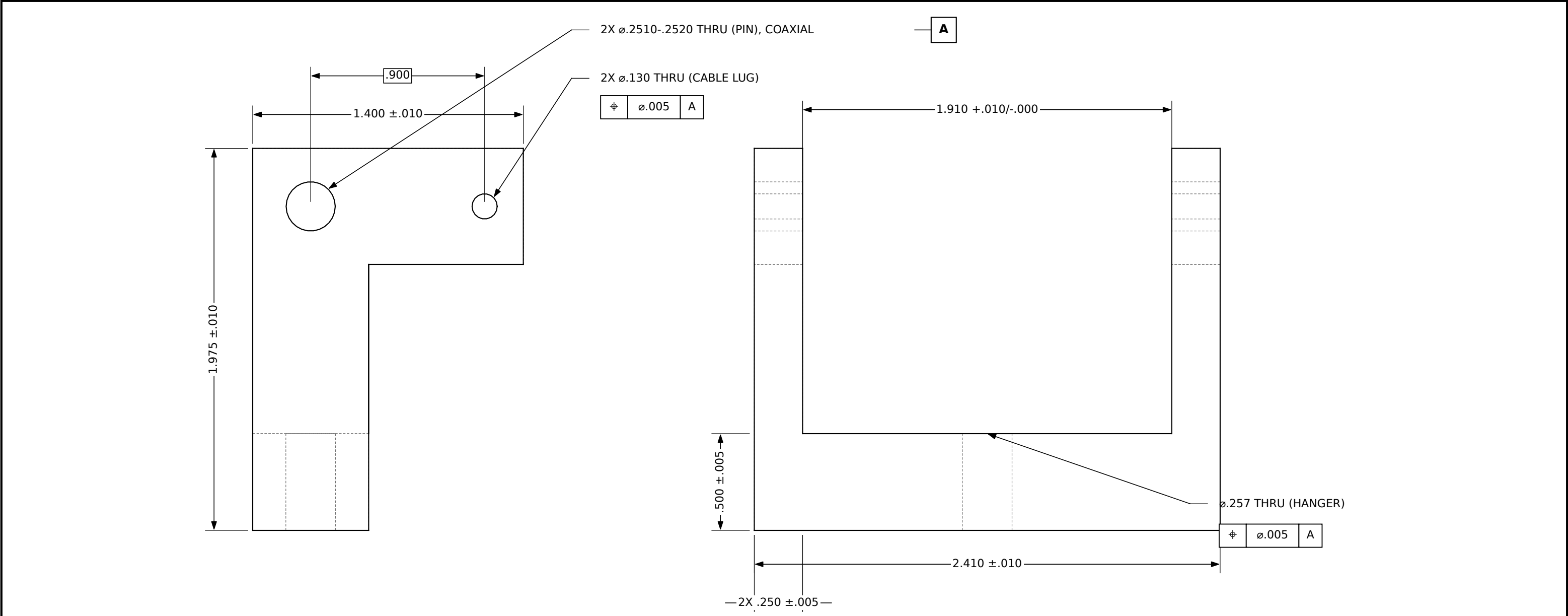






UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: .X ±.03 .XX ±.010 .XXX ±.005 .XXXX ±.0005 ANGLES ±0°30' BREAK SHARP EDGES .005-.015 SURFACE FINISH 63 μin Ra MAX INTERPRET PER ASME Y14.5-2018 UNDIMENSIONED GEOMETRY PER STEP MODEL	
	THIRD ANGLE

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
STIRRUP, 45 DEG CALIBRATION							
MATERIAL				FINISH			
6061-T6 ALUMINUM, .300 PLATE				CLEAR ANODIZE TYPE II			
DWG NO.	REV	SIZE	SCALE	WEIGHT	SHEET		
WT-6C-303	A	B	1.5:1 / 4:1	0.09 lb	16 OF 35		



FRONT VIEW 2:1

VIEW FROM FRONT (-X) 2:1

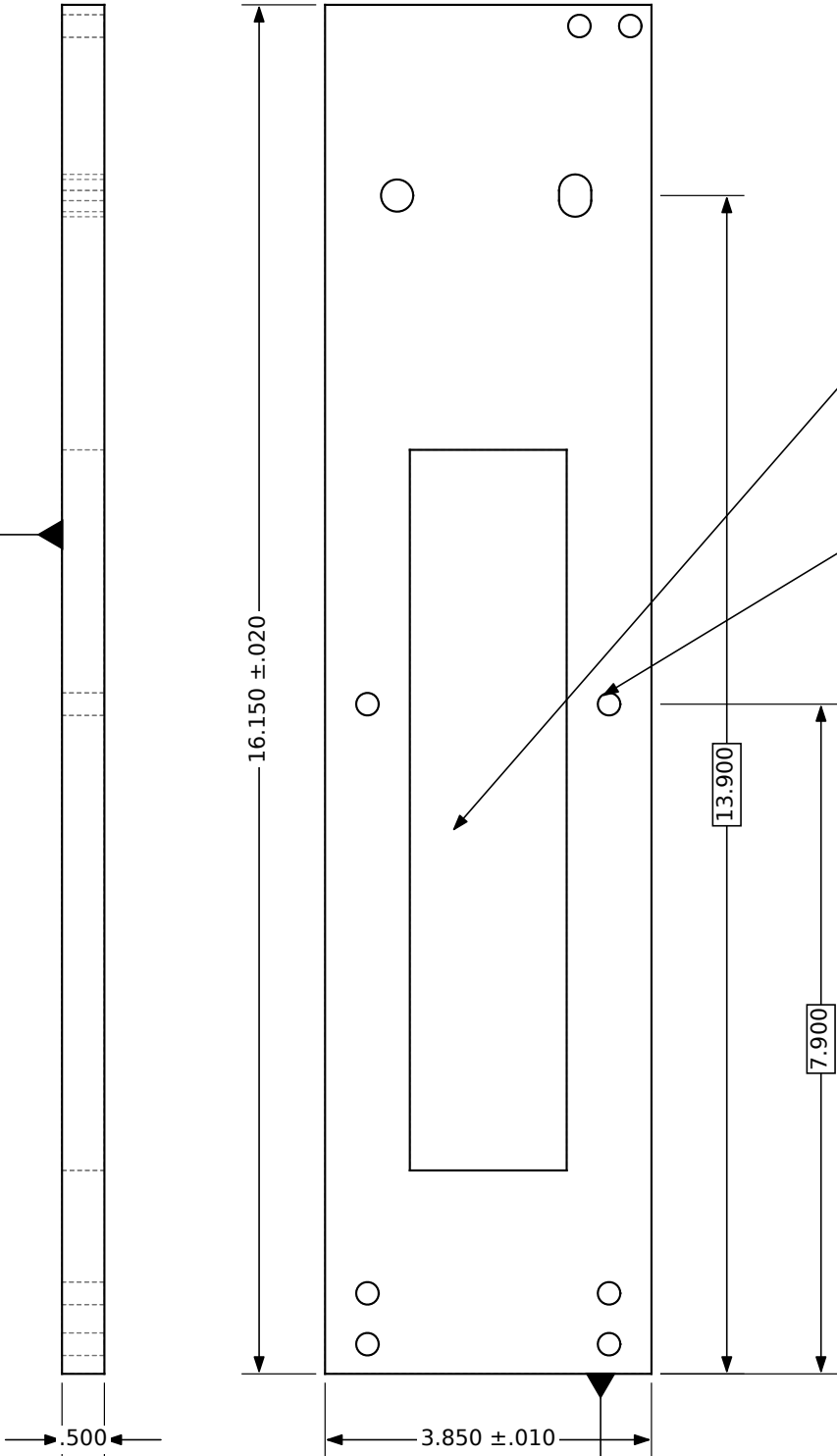
- NOTES:**
- 1. MAY BE MADE AS 3 PIECES (2 STRAPS + BAR) JOINED WITH 2X #8-32 SHCS + 1/8 DOWEL PER JOINT; LINE-REAM THE PIN HOLES AFTER ASSEMBLY.
 - 2. HANGS ON A 1/4 GROUND PIN THROUGH THE SLEEVE STATION HOLES; ITS WEIGHT ACTS AT THE PIN AXIS. WEIGH THE FINISHED YOKE AND UPDATE analysis/calibration_6c_mass_properties.json.
 - 3. SAME YOKE FOR ROLL 0 / 90 / 180 / 270 DEG (ON WHICHEVER PAIR OF SLEEVE WALLS IS HORIZONTAL).
 - 4. CABLE LUGS TAKE THE DRAG CABLE (LAB +X) AT THE SAME STATION.

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: .X ±.03 .XX ±.010 .XXX ±.005 .XXXX ±.0005 ANGLES ±0°30' BREAK SHARP EDGES .005-.015 SURFACE FINISH 63 µin Ra MAX INTERPRET PER ASME Y14.5-2018 UNDIMENSIONED GEOMETRY PER STEP MODEL  THIRD ANGLE	PROJECT				DRAWN / DATE			
	Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
	TITLE				CHECKED		APPROVED	
	YOKE, CALIBRATION, 6-COMPONENT							
	MATERIAL				FINISH			
6061-T6 ALUMINUM				CLEAR ANODIZE TYPE II				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-304		A	B	2:1		0.13 lb		17 OF 35

NEXT ASSY: WT-6C-300 CALIBRATION ASSY

YOKE, CALIBRATION, 6-COMPONENT

- NOTES:**
1. QTY 2, IDENTICAL (ONE EACH SIDE OF THE ROLL HOUSING).
 2. PIVOT HOLE AND LEVELING SLOT TAKE PLAIN 3/8 STUDS IN THE ROLL HOUSING CAM-FOLLOWER HOLES (WT-6C-202); THE SLOT GIVES +-1.6 DEG OF LEVELING.
 3. DATUM A = INNER FACE: BOLTS AGAINST THE BASE TUBE SIDE WALL (WT-6C-307), .050 GAP TO THE HOUSING.
 4. MACHINE THE PIVOT HOLE AND SLOT WITH BOTH PLATES STACKED AND PINNED (ONE SETUP).



SIDE VIEW + FRONT VIEW 1:2.2

WINDOW 1.85 x 8.50 THRU
(LIGHTENING)

2X $\varnothing$.266 THRU
(CROSS SPACERS)

$\varnothing$	.010	A	B	C
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UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X $\pm$.03 .XX $\pm$.010

.XXX $\pm$.005 .XXXX $\pm$.0005

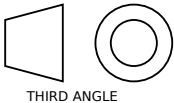
ANGLES $\pm$ 0°30'

BREAK SHARP EDGES .005-.015

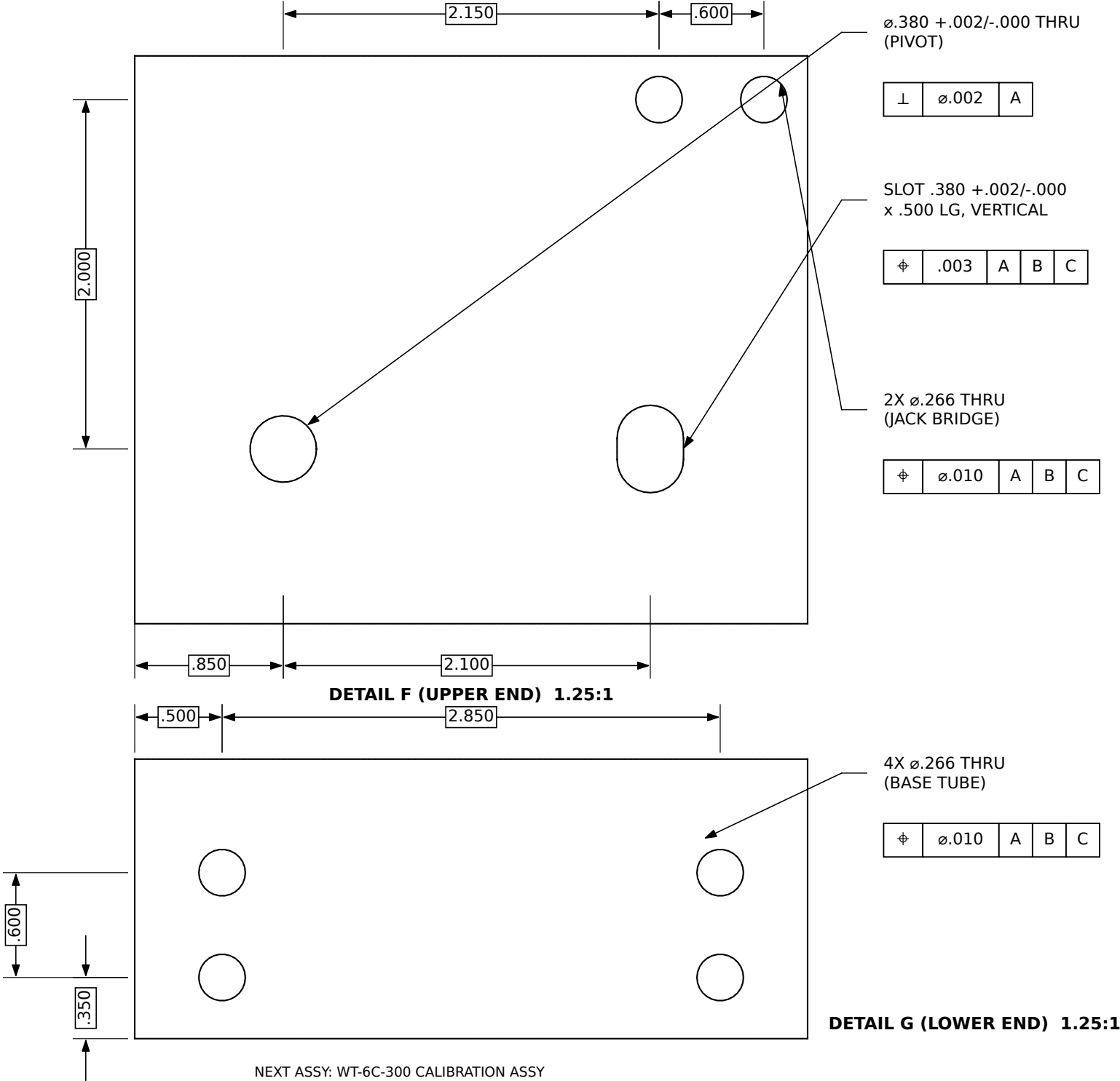
SURFACE FINISH 63 μ in Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



THIRD ANGLE



$\varnothing$.380 +.002/-0.000 THRU
(PIVOT)

$\perp$	$\varnothing$.002	A
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SLOT .380 +.002/-0.000
x .500 LG, VERTICAL

$\varnothing$	.003	A	B	C
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2X $\varnothing$.266 THRU
(JACK BRIDGE)

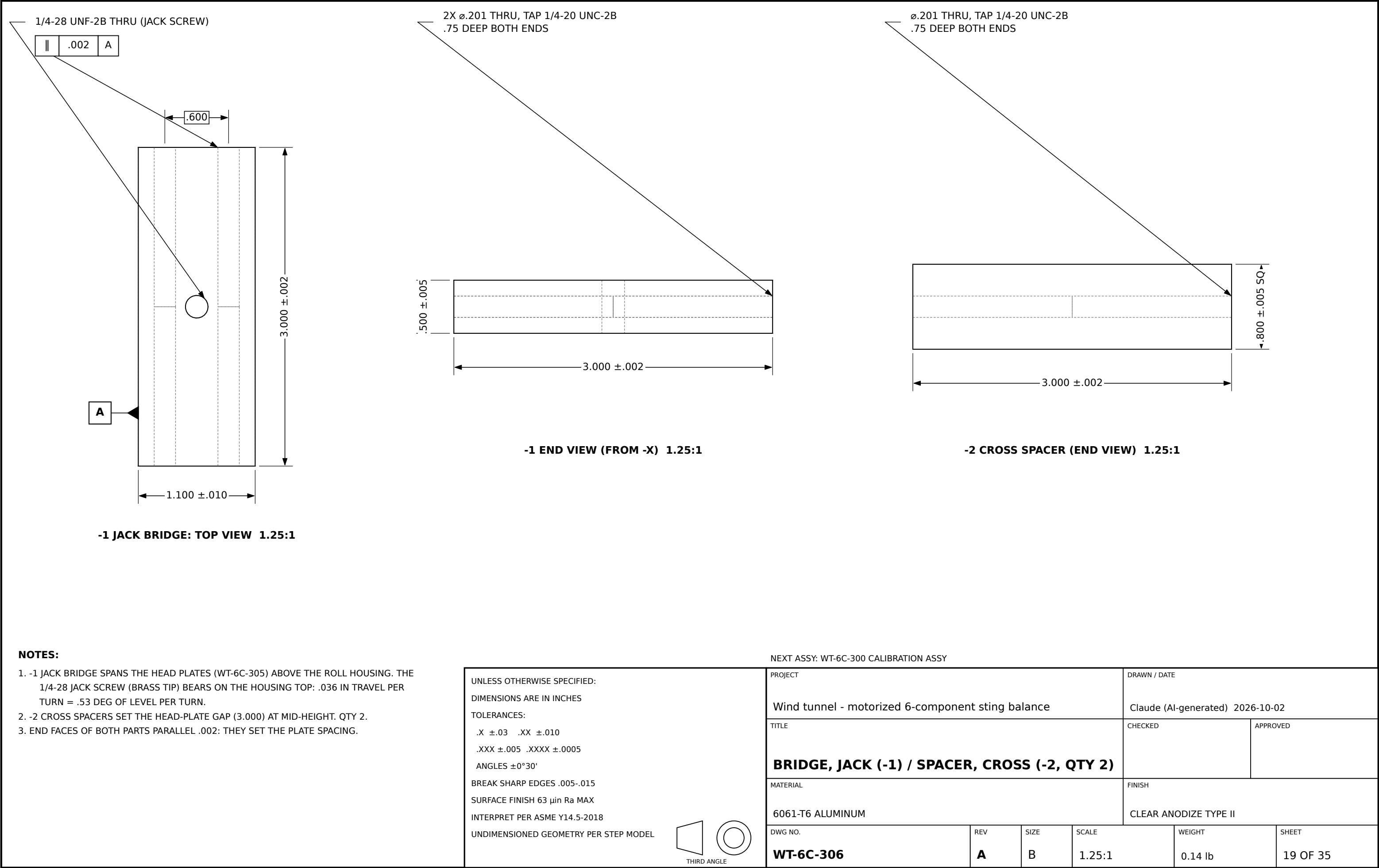
$\varnothing$	.010	A	B	C
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4X $\varnothing$.266 THRU
(BASE TUBE)

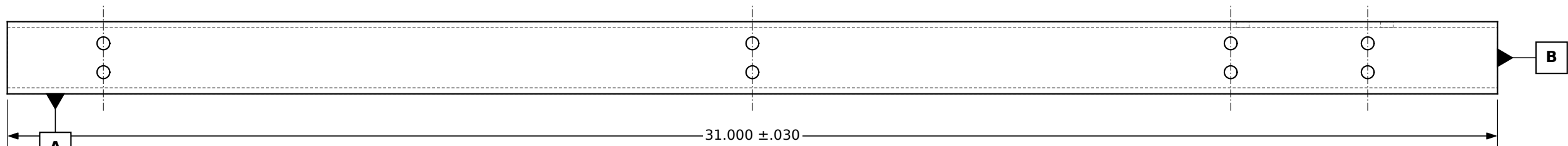
$\varnothing$	.010	A	B	C
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NEXT ASSY: WT-6C-300 CALIBRATION ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
PLATE, HEAD, CALIBRATION STAND (QTY 2)								
MATERIAL				FINISH				
6061-T6 ALUMINUM, .500 PLATE				CLEAR ANODIZE TYPE II				
DWG NO.	REV	SIZE	SCALE	WEIGHT	SHEET			
WT-6C-305	A	B	1:2.2 / 1.25:1	2.23 lb	18 OF 35			



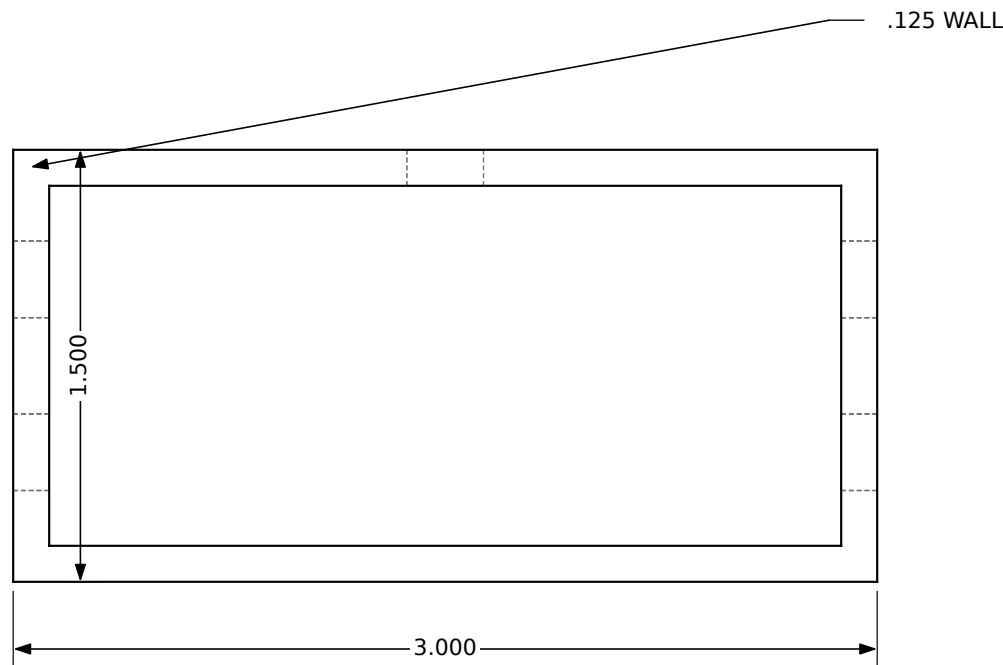
A



FRONT VIEW 1:2.5

HOLE TABLE - ALL Ø.266 THRU; POSITION Ø.010 A B (BASIC DIMS)

HOLES	WALL	FROM B (X)	FROM A (Z)	PURPOSE
H1-H4	SIDE WALLS THRU	29.000	.450 / 1.050	THRUST UPRIGHT
H5-H8	SIDE WALLS THRU	15.500	.450 / 1.050	DRAG UPRIGHT
H9-H12	SIDE WALLS THRU	5.550	.450 / 1.050	HEAD PLATE (NEW)
H13-H16	SIDE WALLS THRU	2.700	.450 / 1.050	HEAD PLATE (NEW)
H17	TOP WALL	5.300	-	3-COMP FOOT BLOCK (OPTIONAL)
H18	TOP WALL	2.300	-	3-COMP FOOT BLOCK (OPTIONAL)



END VIEW 1.5:1

NOTES:

1. CUT TO LENGTH, ENDS SQUARE .005. ALL HOLES Ø.266 (1/4 BOLTS). SIDE-WALL HOLES DRILLED THROUGH BOTH WALLS IN ONE PASS.
2. HOLE TABLE DIMENSIONS FROM THE AFT END (DATUM B) AND THE BOTTOM FACE (DATUM A).
3. PULLEY UPRIGHT HOLES (DRAG $x = 7.0$, THRUST $x = -6.5$ FROM THE BMC) ARE UNCHANGED FROM THE 3-COMPONENT STAND; HEAD-PLATE HOLES ARE NEW FOR THE 6-COMPONENT HEAD.
4. CLAMP TO THE BENCH WITH 4 C-CLAMPS OR BOLT THROUGH THE BOTTOM WALL; LEVEL THE TOP WITHIN .02 DEG.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X $\pm$.03 .XX $\pm$.010

.XXX ±.005 .XXXX ±.0005

ANGLES $\pm 0^{\circ}30'$

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 $\mu\text{in Ra MAX}$

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



THIRD ANGLE

NEXT ASSY: WT-6C-300 CALIBRATION ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
TUBE, BASE, CALIBRATION STAND							
MATERIAL				FINISH			
6061-T6 RECT. TUBE 3.00 x 1.50 x .125 WALL				AS SUPPLIED; DEBURR			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-307		A	B	1:2.5 / 1.5:1		3.20 lb	20 OF 35

8

7

6

5

4

3

2

1

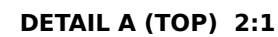
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D

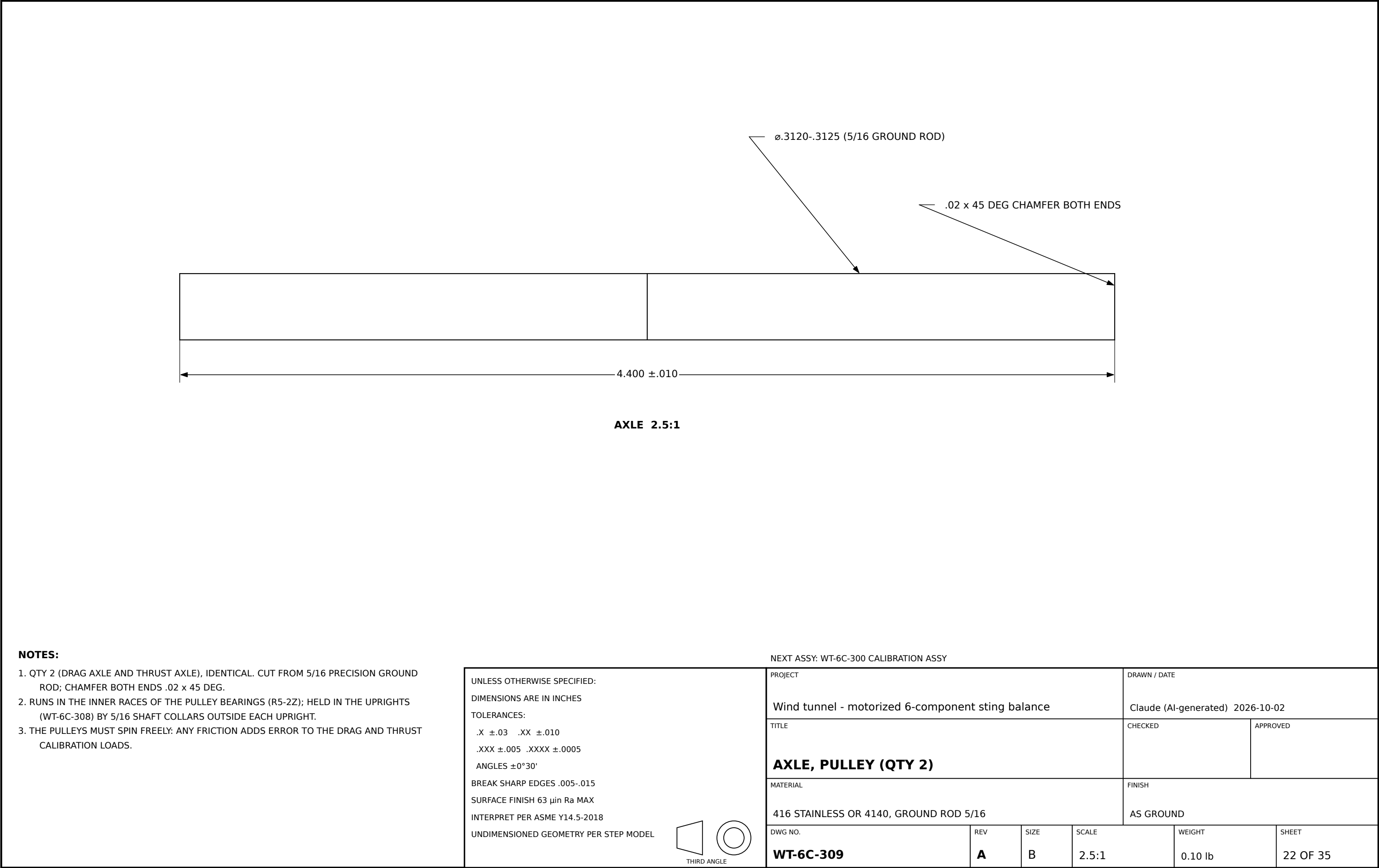
D



1. QTY 4, IDENTICAL: 2 FOR THE DRAG AXLE (X = +7.0 FROM THE BMC), 2 FOR THE THRUST AXLE (X = -6.5).
2. BOLTS TO THE BASE-TUBE SIDE WALLS (WT-6C-307) WITH 2X 1/4-20 SHCS + NUTS.
3. THE AXLE HOLE HEIGHT SETS THE CABLE LINE ON THE BALANCE AXIS: KEEP 12.950 BASIC FROM THE BOTTOM EDGE.
4. MACHINE ALL FOUR STACKED SO THE AXLE HOLES LINE UP.



PROJECT					DRAWN / DATE		
Wind tunnel - motorized 6-component sting balance					Claude (AI-generated) 2026-10-02		
TITLE					CHECKED		APPROVED
UPRIGHT, PULLEY (QTY 4)							
MATERIAL					FINISH		
6061-T6 ALUMINUM, .375 PLATE					CLEAR ANODIZE TYPE II		
DWG NO.			REV	SIZE	SCALE	WEIGHT	SHEET
WT-6C-308			A	B	1:2 / 2:1	0.74 lb	21 OF 35



HOLE TABLE (BASIC, FROM THE PITCH AXIS) POSITION $\emptyset.005$ A B C

HOLE	SIZE	X	Z
H1	ø.281 THRU (TIE)	17.353	4.391
H2	ø.281 THRU (TIE)	18.710	4.734
C	ø.281 THRU (SPACER)	14.600	-8.600
H4	ø.281 THRU (SPACER)	17.000	-8.650
H5	1/4-20 x .75 DP (EDGE)	14.200	-9.100
H6	1/4-20 x .75 DP (EDGE)	15.600	-9.100
H7	1/4-20 x .75 DP (EDGE)	17.400	-9.100

$\oplus$	$\varnothing.005$	A	B	C
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2X ARC SLOT .752 $+.002/-0.000$ WIDE
FULL R ENDS

$\frac{1}{2}$	.002	A	B	C
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PITCH AXIS (BMC)
CENTRE OF ALL ARCS

FRONT VIEW 1:2.2

SIDE

NOTES:

1. QTY 2, IDENTICAL (+Y AND -Y). PROFILE AND SLOTS: CNC MILL (OR WATERJET + FINISH MILL THE SLOTS).
2. THE TWO ARC SLOTS ARE THE CAM-FOLLOWER TRACKS: THEIR CENTRE IS THE PITCH AXIS (BMC). MACHINE BOTH CHEEKS STACKED AND PINNED IN ONE SETUP. SLOT FACES Ra 32 μ m, NO STEPS AT TOOL ENTRY.
3. PITCH-AXIS ORIGIN: 9.100 ABOVE DATUM B AND 14.600 FORWARD OF HOLE C (BASIC). HOLE TABLE COORDINATES ARE FROM THE PITCH AXIS, X AFT, Z UP.
4. BOTTOM EDGE: 3X 1/4-20 UNC-2B .75 DEEP FOR THE BASE PLATE BOLTS (WT-6C-403).

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X $\pm$.03 .XX $\pm$.010

.XXX ±.005 .XXXX ±.0005

ANGLES $\pm 0^{\circ}30'$

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 $\mu\text{in Ra MAX}$

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL

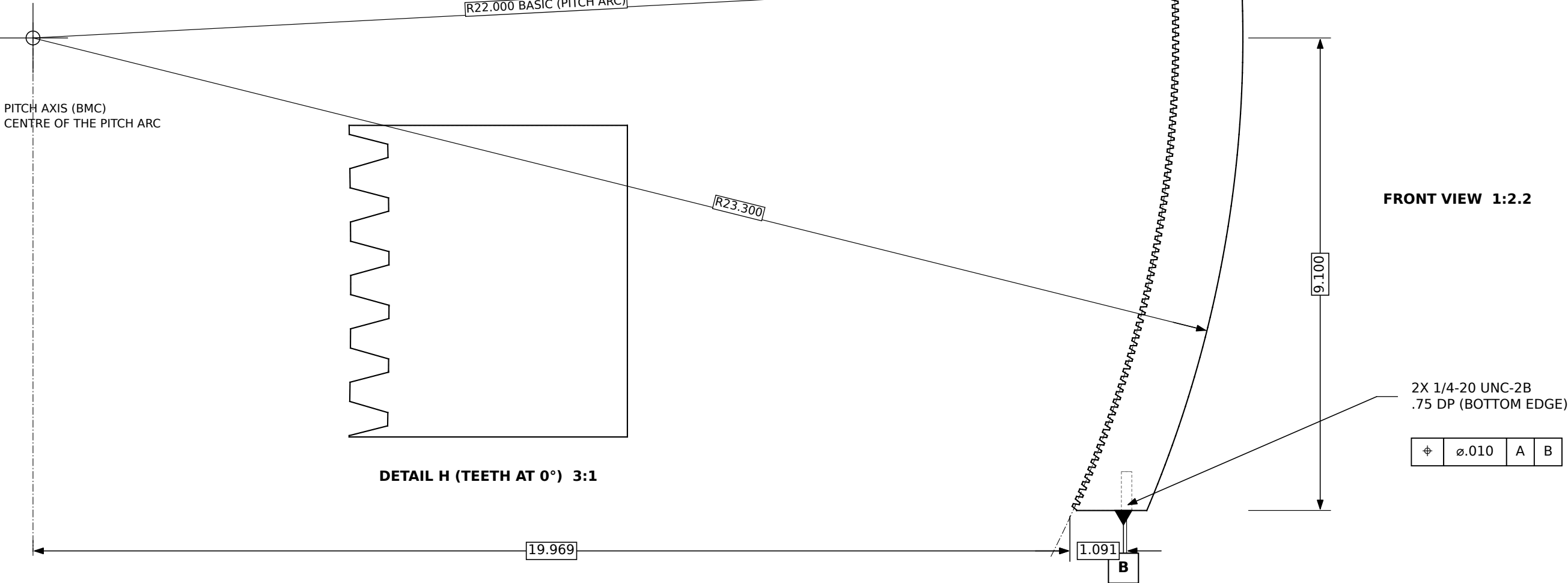


NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT					DRAWN / DATE		
Wind tunnel - motorized 6-component sting balance					Claude (AI-generated) 2026-10-02		
TITLE					CHECKED		APPROVED
CHEEK, PITCH SECTOR (QTY 2)							
MATERIAL					FINISH		
6061-T6 ALUMINUM, .500 PLATE					CLEAR ANODIZE TYPE II		
DWG NO.		REV	SIZE	SCALE	WEIGHT		SHEET
WT-6C-401		A	B	1:2.2	2.59 lb		24 OF 35

GEAR DATA

TYPE	INTERNAL ARC RACK (SEGMENT)
MODULE	1.25 mm
PRESSURE ANGLE	20°
PITCH RADIUS	22.000 in (558.8 mm) BASIC
TEETH ON SEGMENT	~87: FROM THE BASE TOP (-24.4°) TO +11°
ANGULAR PITCH	0.4026° BASIC
TOOTH THK AT PITCH	1.96 mm (-0.05/-0.10)
ADDENDUM / DEDENDUM	1.25 / 1.56 mm
MATING PINION	WT-6C-406, 20 T
QUALITY	AGMA Q7 / ISO 9



NOTES:

- TEETH: WIRE EDM (PREFERRED) OR CNC MILL WITH A MODULE 1.25 FORM CUTTER; PROFILE AND PROFILE FROM THE GEAR DATA TABLE. DEBURR TOOTH ENDS.
- THE PITCH ARC IS CENTRED ON THE PITCH AXIS (BMC), SAME ORIGIN AS THE CHEEKS (WT-6C-401): 9.100 ABOVE DATUM B.
- MATES THE 20-TOOTH PINION WT-6C-406. GRAVITY ALWAYS LOADS THE SAME TOOTH FLANK, SO BACKLASH DOES NOT REVERSE.
- FOOT: 2X 1/4-20 UNC-2B .75 DEEP INTO THE BOTTOM EDGE FOR THE BASE PLATE (WT-6C-403).

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



THIRD ANGLE

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
RACK PLATE, PITCH (INTERNAL ARC RACK)								
MATERIAL				FINISH				
6061-T6 ALUMINUM, .500 PLATE				CLEAR ANODIZE TYPE II (TEETH INCLUDED)				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-402		A	B	1:2.2 / 3:1		0.89 lb		25 OF 35

